Proceedings of the 4th International Conference on Automation, Computing and Renewable Systems (ICACRS-2025)
IEEE Xplore Part number: CFP25CB5-ART; ISBN: 979-8-3315-4886-5

Smart Wildlife Detection and Repulsion System using Deep-Learning

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***Abstract*— Recent advancement in Deep Learning and Data collection have enabled the precise real-time animal recognition and Species-specific ultrasonic repellence for preventing vehicle collisions, livestock predation, and crop damage. The Animal detection is detected by the Animal Feature coding system and detected using the YOLOv5, Tiny YOLOv5 and DNN Preprocessing, Feature Extraction and Classification Techniques. Repel the detected animals with ultrasonic sound, which is species- specific using Arduino board and ultrasonic emitter. The system is developed for real-time animal detection and repellence, so the possibility of vehicle collisions, livestock predation and crop damage are minimized by the advanced technology.**

***Keywords*— Animal detection, Ultrasonic emission, Repulsion.**

I. Introduction

Human-wildlife conflict is more challenging and deeply worrying. It is due to the increasing human population entering wildlife zones. Wildlife is suffering from urban sprawl, agricultural expansion and infrastructure development, all of which can lead to human-wildlife conflict (HWC). HWC result in negative outcomes for either or both species. These conflicts consist of accidents involving animals and vehicles, destroying crops and leading to a loss of biodiversity[11]. To solve conflicts between humans and wildlife an automatic system capable of real time detection and repel the animal with a species specific ultrasonic sound emission is developed. With the help of object detection systems like YOLOv5 and Tiny YOLOv5, the system can able to track and classify animals as they move[15]. To identify different animal species, Deep Neural Networks (DNN) are used for feature extraction and classification. Tensor Flow is used to train the model, OpenCV is used to process images and videos effectively[14]. On the other side, the hardware system is designed with an Arduino microcontroller, ultrasonic sensors and sound emitters to provide species-specific responses in an automated manner (Fig.1). Ultrasonic frequencies is designed to repel the detected animals, ensuring intervention sound are ethical and non-harmful sound[1]. This software and hardware components are efficient, cost-effective and scalable solution aimed to reduce the human-wildlife conflict. The proposed system encourages peace between human and wildlife. It demonstrates how technology provides smart solutions to problems in the

2025 4th International Conference on Automation, Computing and Renewable Systems (ICACRS) | 979-8-3315-4886-5/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICACRS67045.2025.11324419

environment[5].

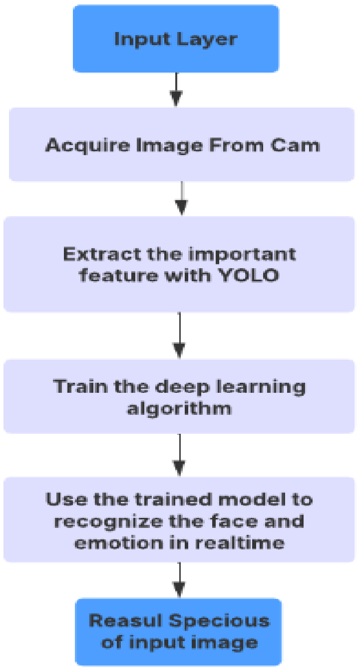

Figure 1 Process chart of the YOLOv5 method

I. METHODOLOGY

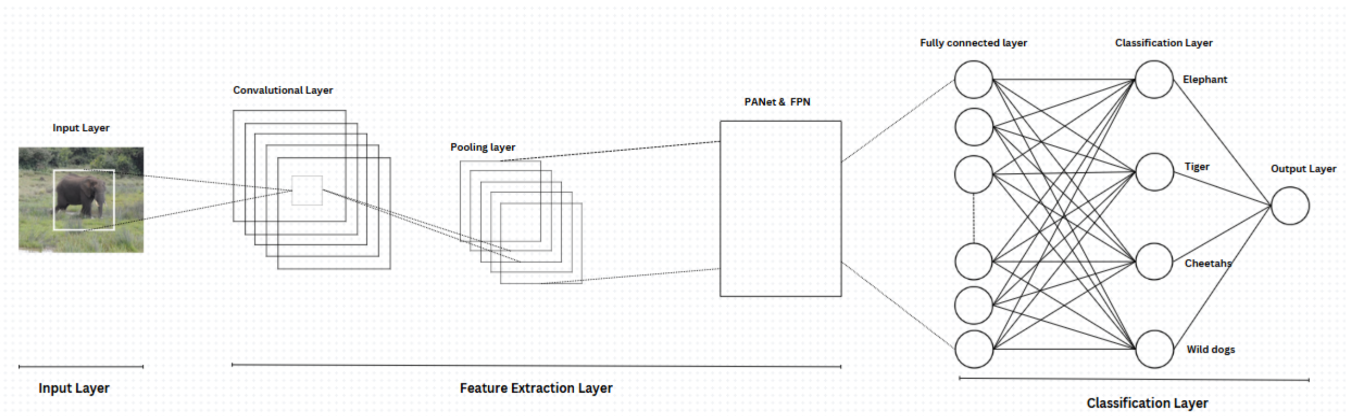

Figure 2 working of YOLOv5 model

Animal detection

In this methodology, we are combining deep learning techniques like DNN, YOLOv5 and TinyYOLOv5 to achieve the animal detection and the Arduino Board, Ultrasonic Sensors and Ultrasonic Emitters are done to repel the detected animal from that region using a species specific fequency [2-4].

The proposed system for animal detection using animal detection uses the DNN to analyze the visual data and it classify animal species accurately. This process will include four stages: Data Acquisition, Preprocessing, Feature extraction , Animal detection [6].

A, Data Acquisition

In this method the Wild-animal images or video are captured using the webcam. This camera focus on the Region of Interest and this includes key features like body shape, movement patterns and unique charaterisrics. This process will be collected and captures the dynamic behaviors effectively and accurate animal detection[7-10].

B, Preprocessing

To provide the Consistent and Reliable input for the feature extraction, the following preprocessing steps are inclused: This may include some process like Animal detection, Segmentation, Normalization and Noise reduction to enhance the quality of the image.

The DNN (YOLOv5 and Tiny YOLOv5) use multiple layers for the feature extraction and classification, that will be the training mechanism for the deep learning frame work. They will train the model using convolution layers, pooling layers and PANet & FPN layers. They will classisies the edges and make the more appropriate image detection mechanism and shape the image needed to detect and we can use the deep learning techniques to make the correct prediction and make the prediction more easier.

Figure 3 Real time Animal Detection using YOLOv5

- **Animal detection:** The YOLOv5 and Tiny YOLOv5 algorithm detects the animals in each of the video frame.
- **Segmentation:** These detected animals are cropped to isolate the Region of Interest (ROI) for the analysis.
- **Normalization:** In this method the animal images are resized to 64×64 pixels to ensure uniformly.
- **Noise Reduction:** This method are used as the filtering techniques that removes the artifacts caused by lighting, blur or the camera noise.

C. Feature Extraction

The preprocessed animal image is analysed by the Deep Neural Network (DNN) represented in Figure 2. YOLOv5 and Tiny YOLOv5 use multiple layers to get accurate animal detection [10-12].

Convolution Layer: These are used to detect the features like edges, textures and patterns that represent the animal characteristics. In this we have used the Convolution Operation for detecting the spatial features like body shape, movement patterns and unique charateristics. The basic mathematical formula for Convolution layer can be represented as in (1).

$$Net_{l,j}= \sum_{i\in M_j} p_{l-1,i} \ . w_{i,j} + q_j \quad (1)$$

Pooling Layer: These will reduce the spatial dimensions by retaining the significant features and we use the pooling operation for the reduction process. The basic mathematical formula for Pooling layer can be represented as in (2).

$$pool_{l,j} = \max\,(a_{l,i}) \quad (2)$$

PANet & FPN:

FPN (Feature Pyramid Network): These are used to enhance the detection of largest and smallest animals by bring out the features at different scales.

PANet (Path Aggregation Network): These will improve information flow between layer and improve the accuracy of animal detection and classification.

Fully Connected Layer: These will combine the extracted features into the feature vector and it will represent the overall animal shape, movement and pattern. The basic mathematical formula for PANet&FPN layer can be represented as in (3).

$$net_l = \sum_i w_i \ . a_i + b \quad (3)$$

D. Animal Classification

In this animal classification the feature vector generated by the DNN(YOLOv5 & Tiny YOLOv5) is transferred to the classification layer and that uses the Softmax function to predict the emotion category and this will classifies the detected animals into categories: Elephant, tigers, cheetahs and wild dogs represented in Figure 3.

E. Advantages of the Method

Real-Time Capability: This mechanism will processes the data efficiently and making it suitable for the real time applications.

Scalability: This method will be extended to large datasets and also could ne integrated into various applications like wildlife monitoring and HWC prevention system.

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Utilizing YOLOv5 for Detection: The YOLOv5 model is integrated for high speed and detecting animal accurately in video frames, enabling fast identification and tracking (Fig.4).

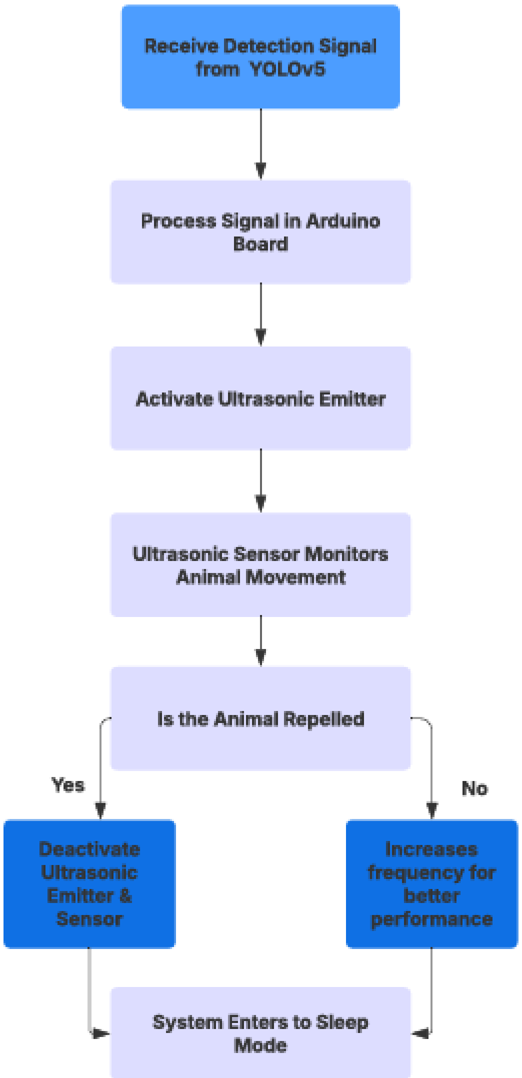

Figure 4 Process chart of the Repulsion System

Animal Repulsion

F. Arduino Board

This will act as the central processing unit(CPU) which receives the signal from the software(YOLOv5 or Tiny YOLOv5) used to detect the animal and then it will activates the ultrasonic repulsion system. Arduino Board will process the detected output and decide the correct frequency of ultrasonic sound to be generated. Arduino Board process is mentioned in Figure 5. It will also ensure power efficiency and operates the system only when required [13].

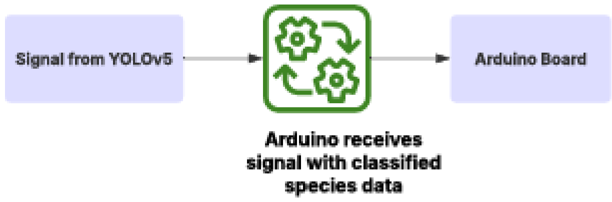

Figure 5 Process of Arduino Board

G. Ultrasonic Sensor

Ultrasonic Sensor are placed in the crucial sites, which will monitor the area continuously to confirm whether the animal is present in the same region after activation of the repulsion system. If the animal remains in the same region the system adjusts the ultrasonic frequency to enhance the effectiveness. Ultrasonic sensor process is mentioned in Figure 6. If the animal repelled from that area, Arduino will deactivates the ultrasonic emitters to save power.

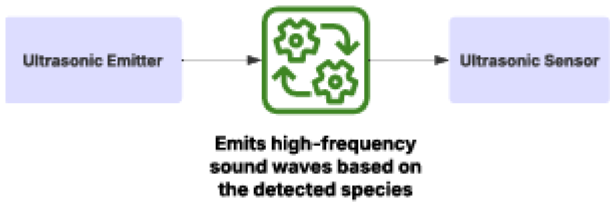

Figure 6 flow of Ultrasonic Senor

H. Ultrasonic Emitter

When an animal is detected the Arduino board receives a detection signal. The Arduino board then activates the ultrasonic emitters, which is used to generate high-frequency sound wave based on the species detected. Different species have different ultrasonic frequency ranges, and based on that, we can repel the animal most effectively. The ultrasonic emitter is placed in the position of covering the entire area and turns off after ensuring the animal has been repelled from that region. Ultrasonic Emitter process is mentioned in Figure 7.

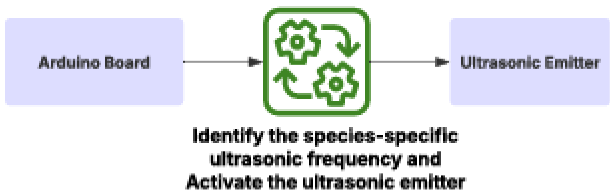

Figure 7 process of Ultrasonic Emitter

I. System Integration for Repulsion

1. **Receiving Detection signal from Software**

The YOLOv5 or Tiny YOLOv5 detects the animal and classifies its species. After detection the classification outcome is transmitted as a digital signal to the Arduino board.

2. Processing the Signal by Arduino Board

The Arduino Board receives and reads the classification data and chooses the correct ultrasonic frequency based on the detected animal species. It will trigger the ultrasonic emitters to generate the ultrasonic frequency and the signal is sent to the ultrasonic sensors to monitor the area continuously.

3. Ultrasonic Emission for Repulsion

The ultra emitters produce high frequency sound waves based on the animal's species, which makes the animal inconvenient and forces it to leave the area.

4. Continuous Monitoring

If the animal remains in the same area, the system adjusts the ultrasonic frequency to enhance the effectiveness. The ultrasonic sensor will track the movement of the animal to ensure the animal has left the place.

5. Ultrasonic emitter and sensor Deactivation

If the animal has leave, the Arduino Board turns off the ultrasonic emitters and sensors to save the power. The system resets to standby mode and waits for the next detection signal.

6. Frequency range for different animal species

- The ultrasonic frequency will be changed by the Arduino board depending on the type of animal is detected, which makes the animal inconvenient and forces it to leave the area.
- Humans are not affected by these ultrasonic sounds because the animal can't hear higher frequencies like humans and the human ear can sense up to 20Hz mentioned in Table I. So, the model effectively repels the detected animal from that area without affecting the people in that area (Fig.8).

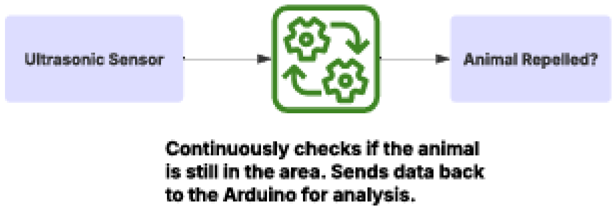

Figure 8 Signal Processing Stage.

Animal Species	Ultrasonic Frequency	Expected Outcome
Elephant	12-15 kHz	Moves away due to discomfort
Tiger	25-30 kHz	Avoid the Area
Cheetah	20-25 kHz	Moves away due to discomfort
Wild Dog	22-28 kHz	Moves away due to discomfort

Table I. Frequency Range

J. Advantages of Repulsion System

Energy-Efficient: The ultrasoinc emitters are activated only if the animal is detected and the signal is sent to the ultrasonic emitters through the Arduino board.

Sleep Mode: When no animal detected the system will enter into the low power mode to save the power and battery.

II. RESULT AND ANALYSIS

The wild-life detection and repulsion System combines deep learning models with an embedded hardware system to detect the animal species and repel it in real-time. The performance of the system is evaluated based on the performance metrics and accuracy metrics, ensuring its effective performance in wildlife conflict resolution. The analysis is to determine the detection speed, accuracy and hardware efficiency over YOLOv5 and Tiny YOLOv5 which are integrated with the hardware systems.

A. Performance Metrics

The performance of YOLOv5 models for animal detection was analysed using the metrics given below.

Recall: It will measures the ability of the model to detect animals. If the recall is higher the missed detection is minimum. The basic mathematical formula for Recall can be represented as in (4).

$$Recall = \frac{TP}{(TP+FN)} \quad (4)$$

Where,

TP refers to the True Positive, which is used to determine if the mode has correctly detected.

FN refers to the False Negatives, which is used to determine if the mode has missed to detect.

Precision: The detected accuracy is evaluated by precision. High precision results the false positive. The basic mathematical formula for Precision can be represented as in (5).

$$Precision = \frac{TP}{TP+FP} \quad (5)$$

Where,

FP refers to the False Positive, which is used to determine if the mode has not detected correctly.

AP & mAP: Area under the Precision - Recall curve is used to calculate detection accuracy. The basic mathematical formula for AP & mAP can be represented as in (6) and (7).

$$AP = \int P(R)\, dr \quad (6)$$

Where,

AP refers to Average Precision.

$P(R)$ is a Precision as a function of Recall.

$$mAP = \frac{1}{K} \sum_{i=1}^{k} AP_i \quad (7)$$

Where,

mAP refers to mean of AP.

N is the number of classes like number of different animal in detection process.

AP_i refers to AP in class i.

IoU: It is used to find the matches between the predicted and actual bounding boxes. The basic mathematical formula for IoU can be represented as in (8).

IoU = Area of Intersection / Area of union (8)

Where,

IoU refers to Intersection over Union.

FPS: Used to indicate the detection speed mainly in real-time application. The basic mathematical formula for FPS can be represented as in (9).

FPS = Total number of Frames / Total time taken (9)

Where,

FPS refers to Frames Per Second.

B. Accuracy Metrics

The detection accuracy of Training dataset, Validation dataset, Confidence score distribution and detection accuracy per image are evaluated. Used to determine the model performance.

1. Train set Detection Accuracy

The Accuracy of train set detection will varies for different images, high detection rate for some set of images and perform loss for the another sets. To improve the consistency it need some dataset. The Accuracy of train set detection is reprencented in Figure 9.

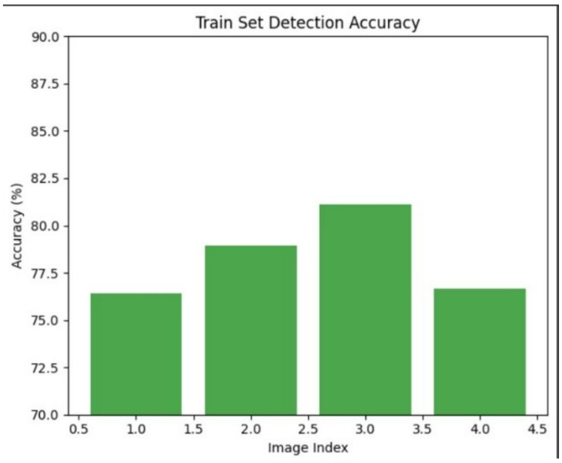

Figure 9 Train Set Detection Accuracy.

Validation Set Detection Accuracy

In the Fig- the Validation set has 80% of accuracy level which shows the good performance of the model in the unseen data. Validation set detection is mentioned in Figure 10.

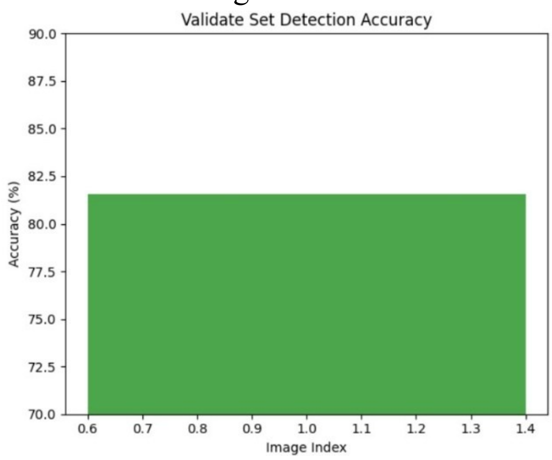

Figure 10 Validation Set Detection Accuracy.

Confidence Score Distribution and Detection .Accuracy Per Image

The Confidence score distribution is maintained with 0.8, which indicate compatible detections or restrictive threshold. While adjusting this threshold refine analysis. The detection accuracy varies from 75% to 85% for different images which is due to the high sun light or background complexity (Fig.11).

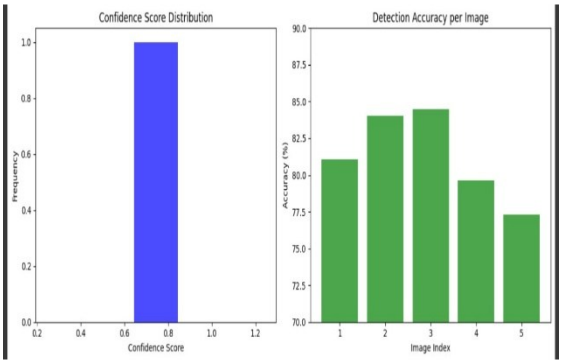

Figure 11 Confidence Score Distribution and Detection Accuracy Per Image.

C. Model Performance

Metric	Training Set	Validation Set
Precision	82.5	80.1
Recall	78.9	76.3
IoU	79.2	77.8
AP	81.4	78.6
mAP	80.7	79.2
FPS	45 FPS	42 FPS
Overall Accuracy	80.3	78.9

Table-2 Performance of the model

III. CONCLUSION

An advanced Intelligent Animal Repelling System designed to safeguard crops against ungulate intrusions using AI, Edge computing, IoT, and LPWAN. This integrates real-time animal detection and repulsion via processes of YOLOv5 and Tiny-YOLOv5 had been tested and optimized for edge computing platforms such as Arduino. A counterintuitive response time, precision, and energy efficiency while ensuring the best operation of the system in rural settings were considered. The study aims at establishing the importance of hardware-software integration, experimental validation, and customized Edge-AI deployment for precision agriculture applications. A backup mechanism involving PIR sensors and cameras ensures activity detection. In case of PIR failure, the camera kicks into detection mode, and the DNN classifies the animal. The ultrasonic repelling device is then activated, and activity data are sent via LoRa to the TTN server. It concludes that although Edge-AI and IoT technologies have substantially advanced, deploying modern applications is still challenging. This solution therefore sets up the equation of possibilities for autonomous wildlife management and further developments in smart agriculture.

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